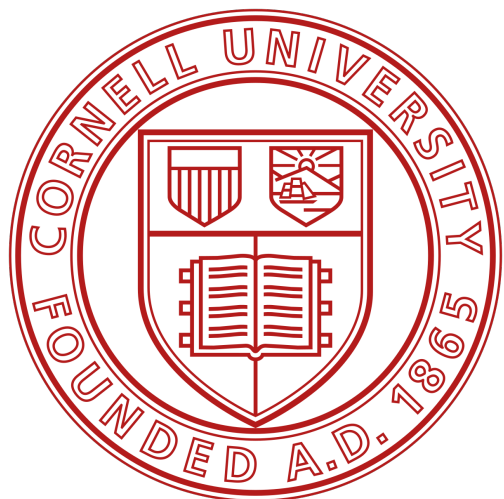


Bluebell: An Alliance of Relational Lifting and Independence For Probabilistic Reasoning

Jialu Bao Emanuele D'Ossualdo Azadeh Farzan



Universität
Konstanz



UNIVERSITY OF
TORONTO



Mar. 13th 2025 Portland State University

Probabilistic Programs

$t ::= \text{skip}$

| $x := e$ (assignment)

| $x \sim \text{flip}(c)$ (sampling)

| $t_1; t_2$ (sequencing)

| if e then t_1 else t_2 (branching)

| repeat e t (bounded loop)

Program Semantics

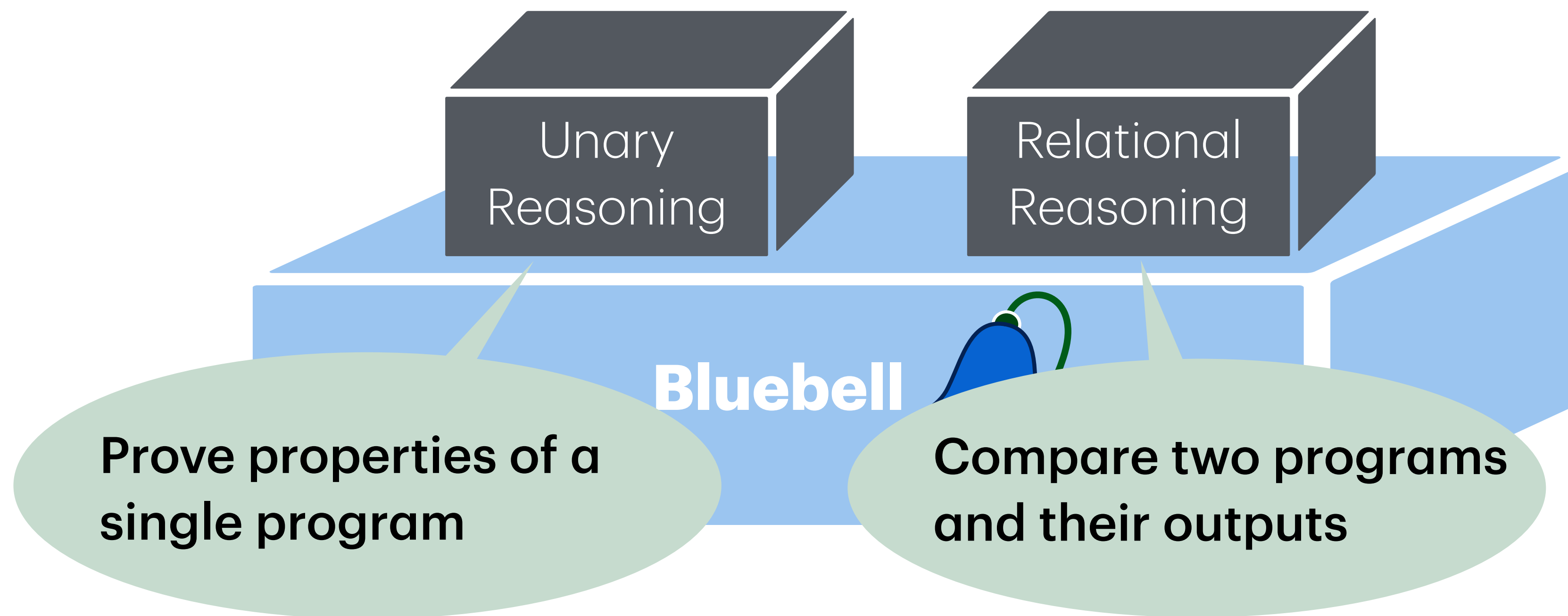
$t ::= \text{skip}$	
$x := e$	(assignment)
$x \sim d$	(sampling)
$t1; t2$	(sequencing)
if e then $t1$ else $t2$	(branching)
repeat e t	(bounded loop)

Program Stores $\mathbb{S} := X \rightarrow V$

Probabilistic Stores $\mathbb{D}(\mathbb{S}) := \mathbb{S} \rightarrow [0,1]$ ✨

Probabilistic Programs $\mathbb{D}(\mathbb{S}) \rightarrow \mathbb{D}(\mathbb{S})$

Deductive Verification of Probabilistic Programs



Unary v.s.
Relational

Why mixing?

How?

Unary Probabilistic Programs

Example. One-Time Pad

```
m :~ flip(0.7);
```

```
k :~ flip(0.5);
```

```
c :~ m xor k
```

c does not leak anything about **m**.

Equivalently, **c** and **m** are independent.

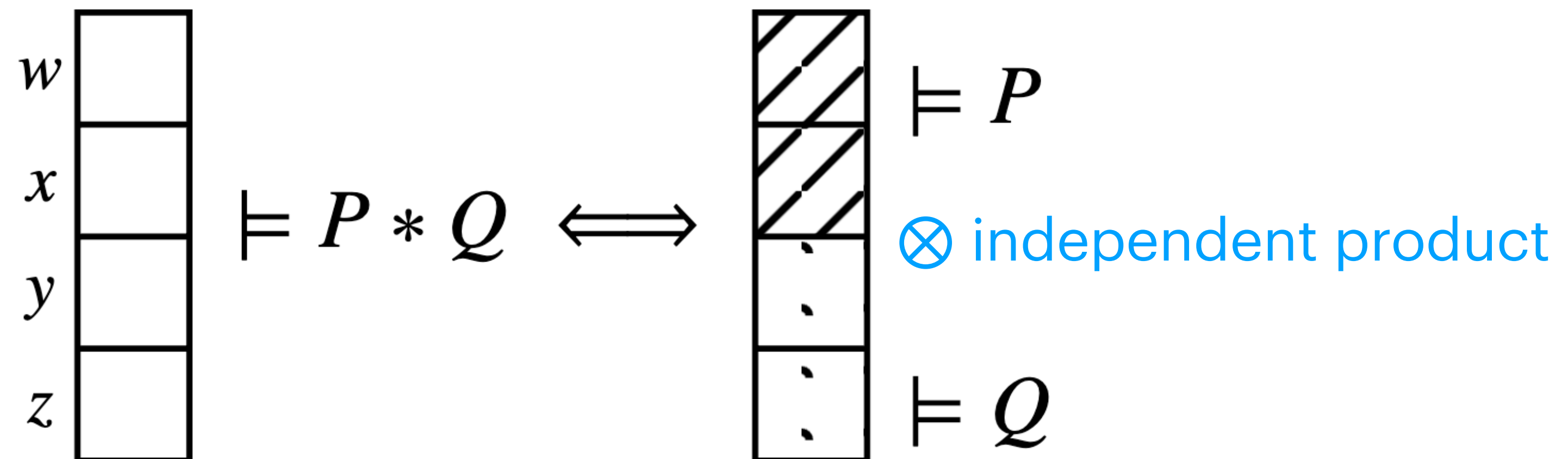
Probabilistic Separation Logic
helps reasoning about
independence

Unary: Probabilistic Separation Logic

- Assertion logic

- $P, Q ::= p \in \mathcal{AP} \mid \top \mid \perp \mid P \wedge Q \mid P \vee Q \mid P \Rightarrow Q \mid P * Q$

- Resource Interpretation



- Program logic

- Judgement: $\{P\}C\{Q\}$

Unary: Probabilistic Separation Logic

Example. One-Time Pad

$\{ T \}$

$m : \sim \text{flip}(0.7);$

$\{ m \sim \text{flip}(0.7) \}$

$\{ \quad \quad T \}$

$k : \sim \text{flip}(0.5);$

$\{ \quad \quad k \sim \text{flip}(0.5) \}$

$c : \sim m \text{ xor } k$

$\{ m \sim \text{flip}(0.7) * c \sim \text{flip}(0.5) \}$

Samp

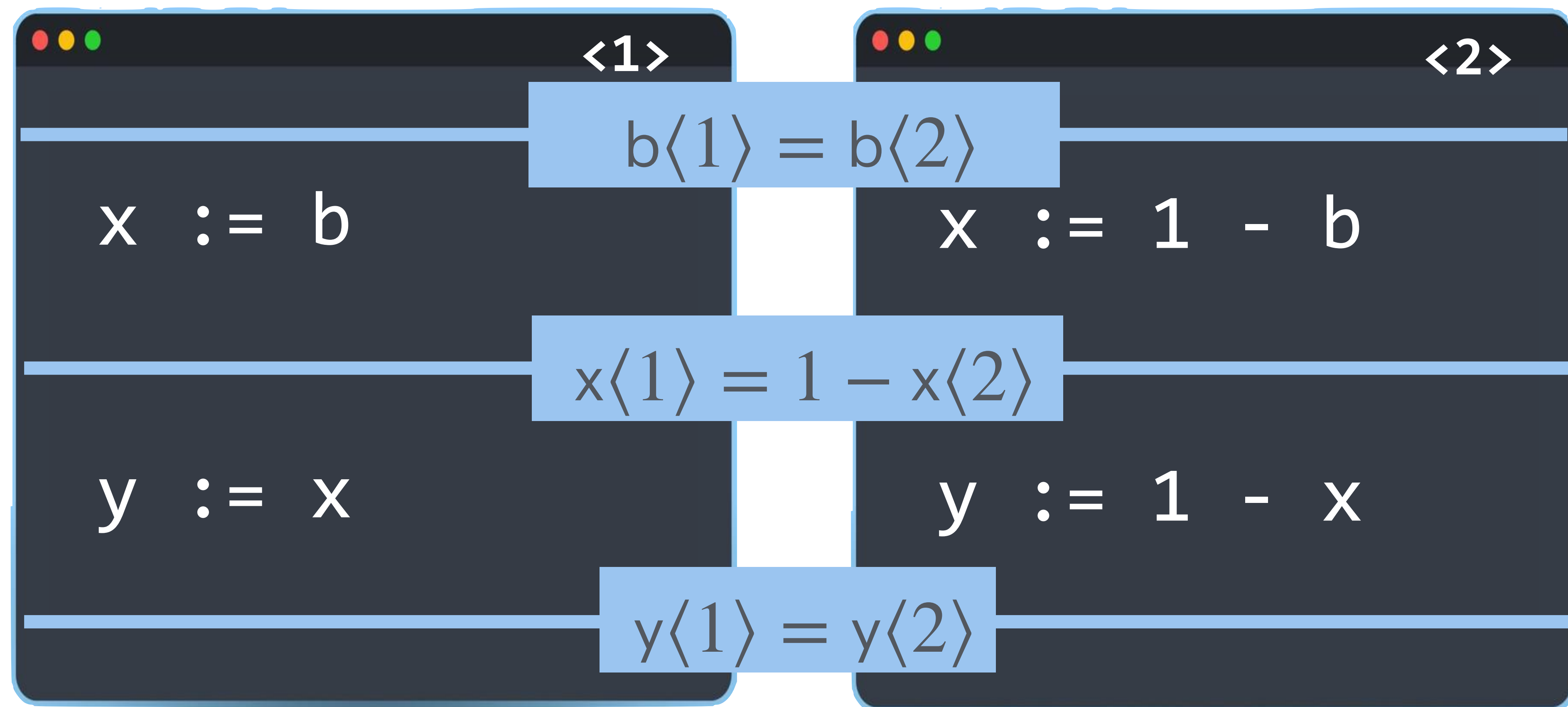
$$\frac{}{\vdash \{ T \} x : \sim \text{flip}(c) \{ x \sim \text{flip}(c) \}}$$

Frame

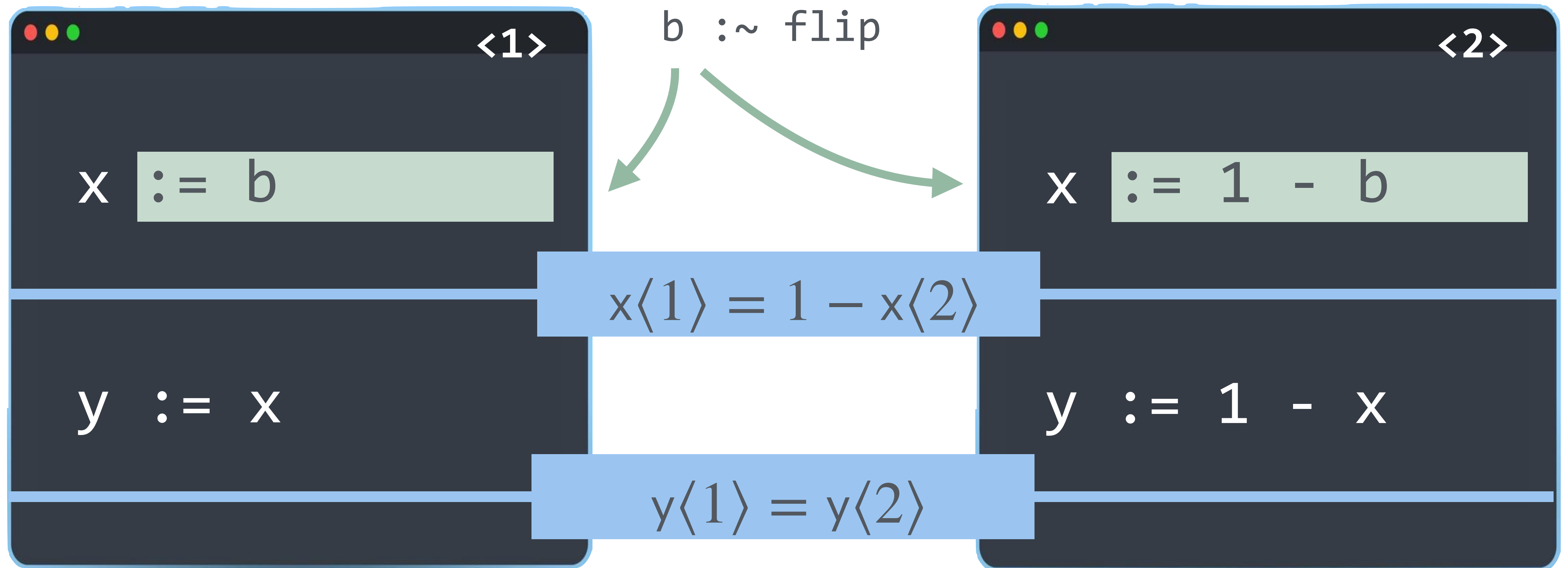
$\frac{\vdash \{ \phi \} c \{ \psi \} \quad \text{side conditions}}{\vdash \{ \phi * \eta \} c \{ \psi * \eta \}}$

Relational Hoare Logic

Proof in lockstep:

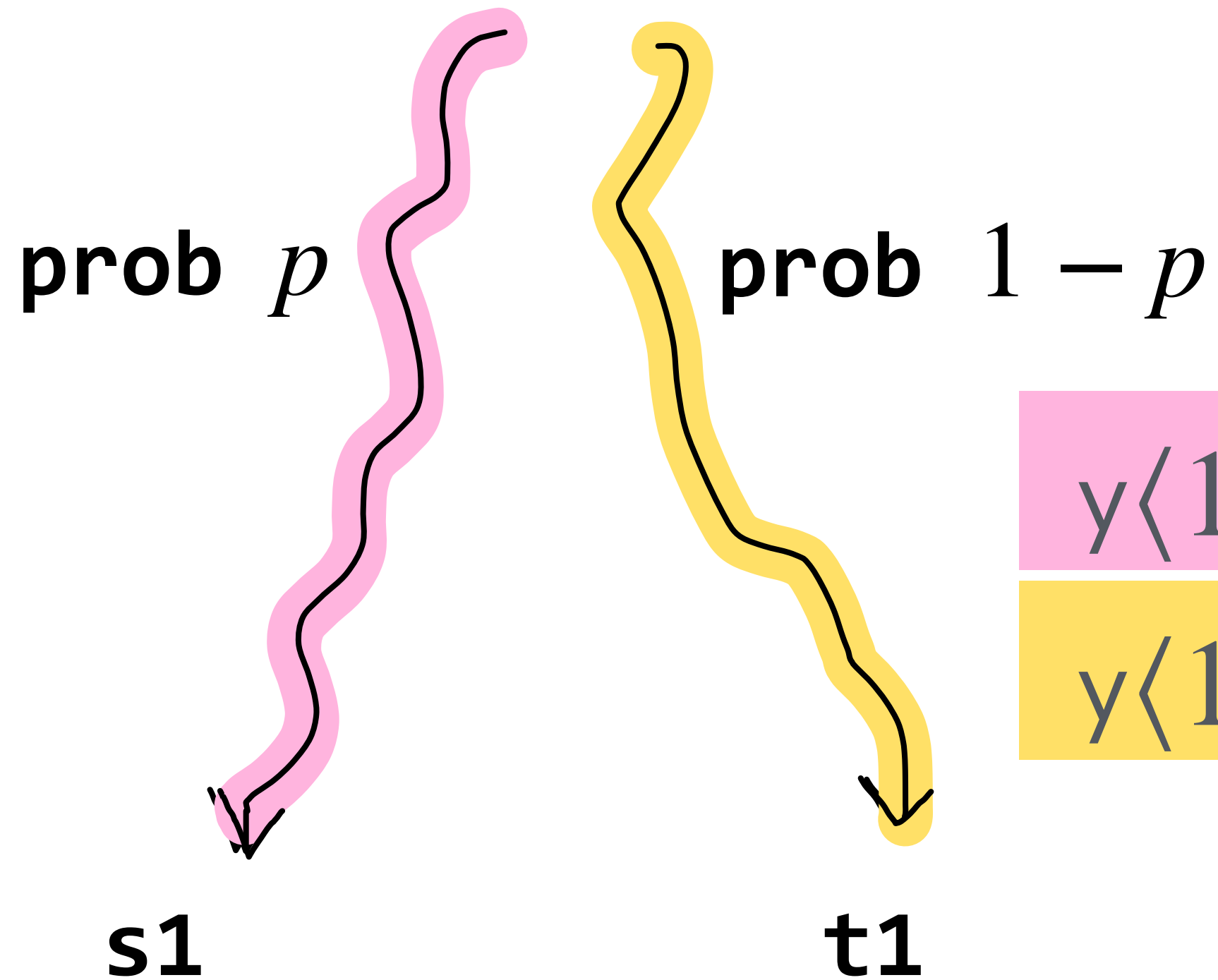


The Coupling Method

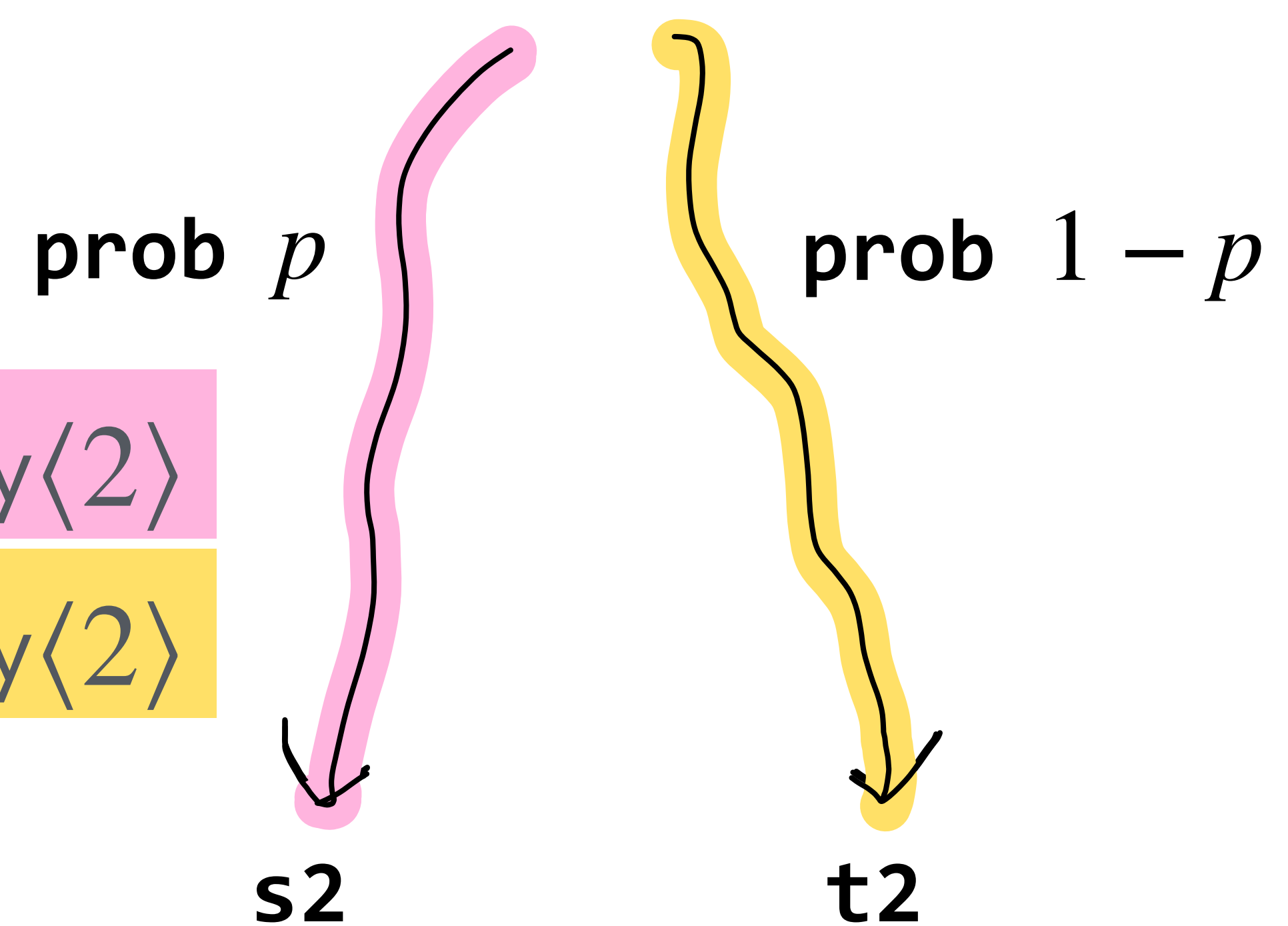


The Coupling Method

Initial State $\langle 1 \rangle$



Initial State $\langle 2 \rangle$

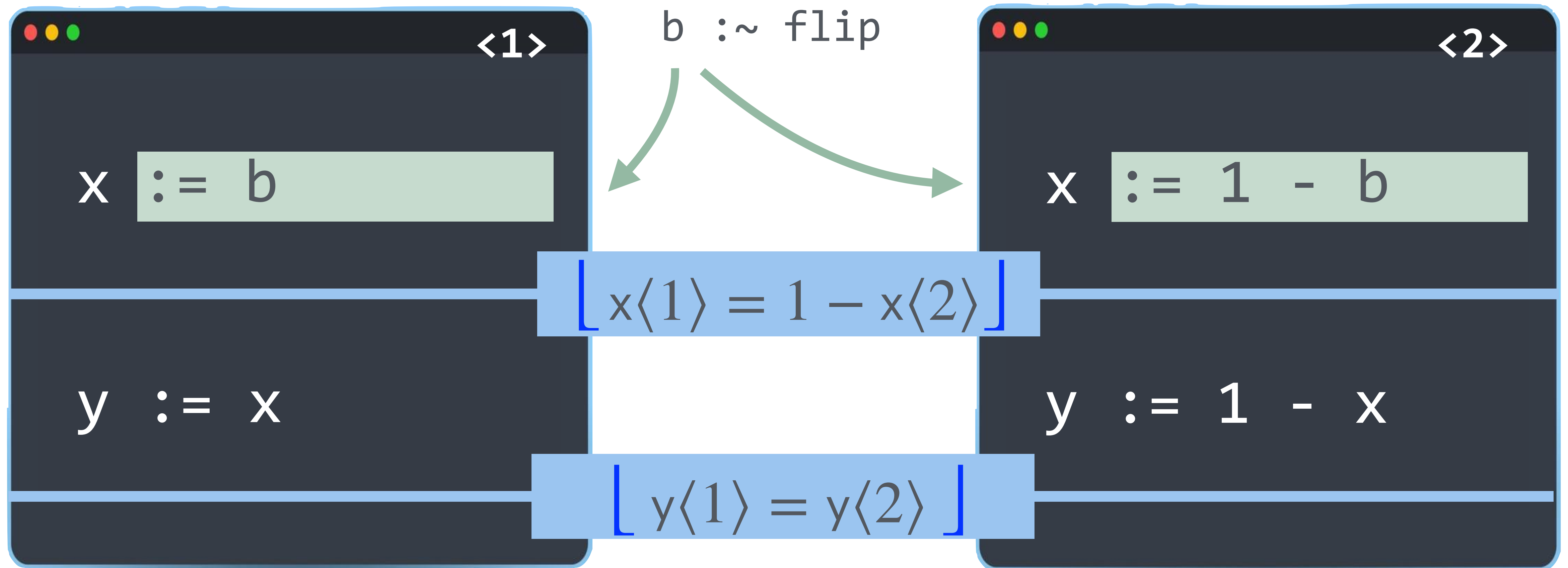


$$y\langle 1 \rangle = y\langle 2 \rangle$$

$$y\langle 1 \rangle = y\langle 2 \rangle$$

Relational Lifting $[y\langle 1 \rangle = y\langle 2 \rangle]$

The Coupling Method



Distribution of $y\langle 1 \rangle$ = Distribution of $y\langle 2 \rangle$

Missed opportunity if
you don't mix them!

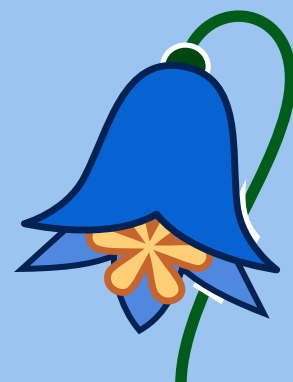
Unary
Reasoning



Relational
Reasoning

[R]

Bluebell



Unary v.s.
Relational

Why mixing?

How?

x and y are independent.

<1>

```
x :~ flip(0.5)  
y :~ flip(0.7)
```

x and y are independent.

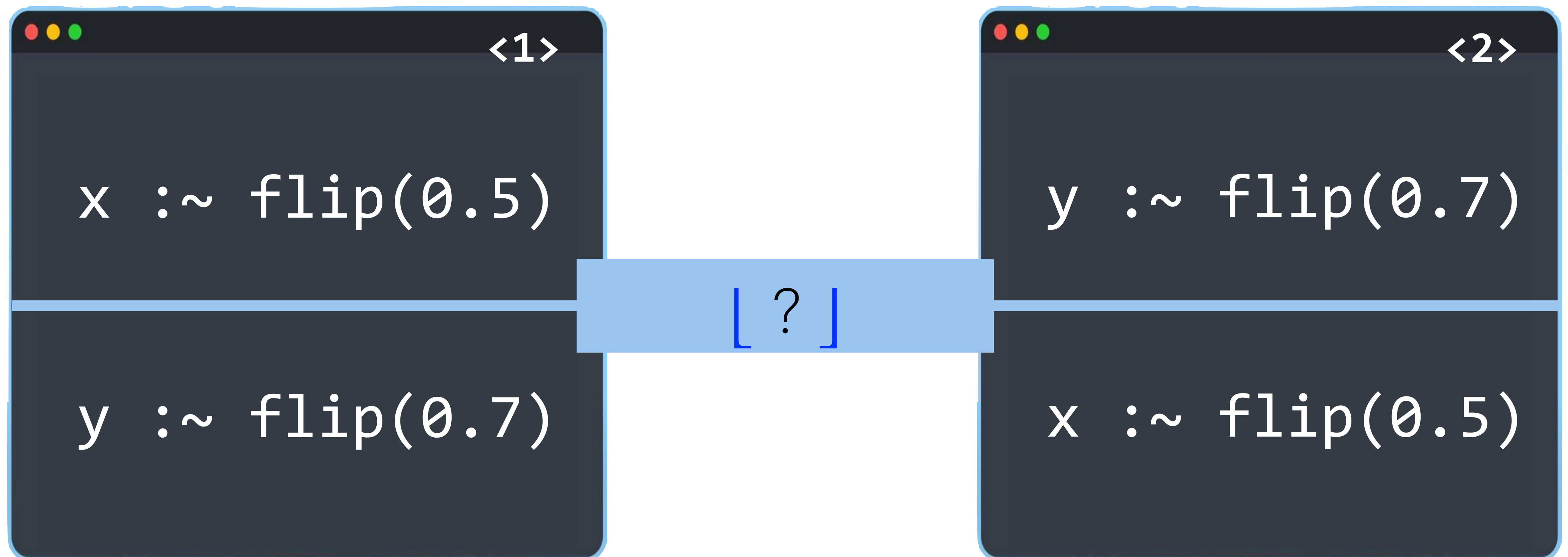
<2>

```
y :~ flip(0.7)  
x :~ flip(0.5)
```

Establish $[x\langle 1 \rangle = x\langle 2 \rangle \wedge y\langle 1 \rangle = y\langle 2 \rangle]$?

Output distribution of program <1> = Output distribution of program <2>

Coupling Method Requires Proving in Lockstep:



Coupling Method Requires Proving in Lockstep.

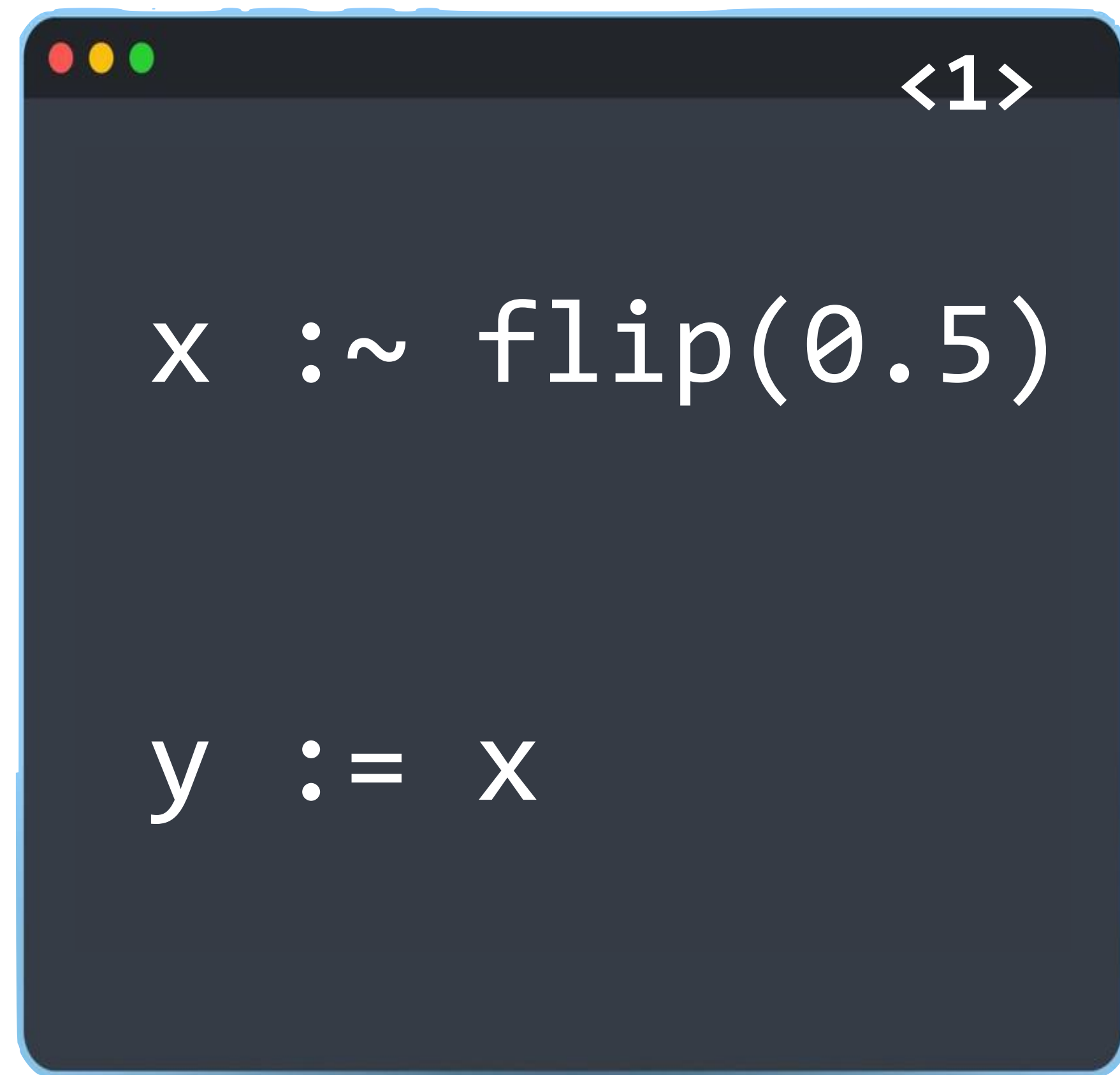
```
<1>  
  
x :~ flip(0.5)  
  
y :~ flip(0.7)  
  
^
```

$[y\langle 1 \rangle = y\langle 2 \rangle]$

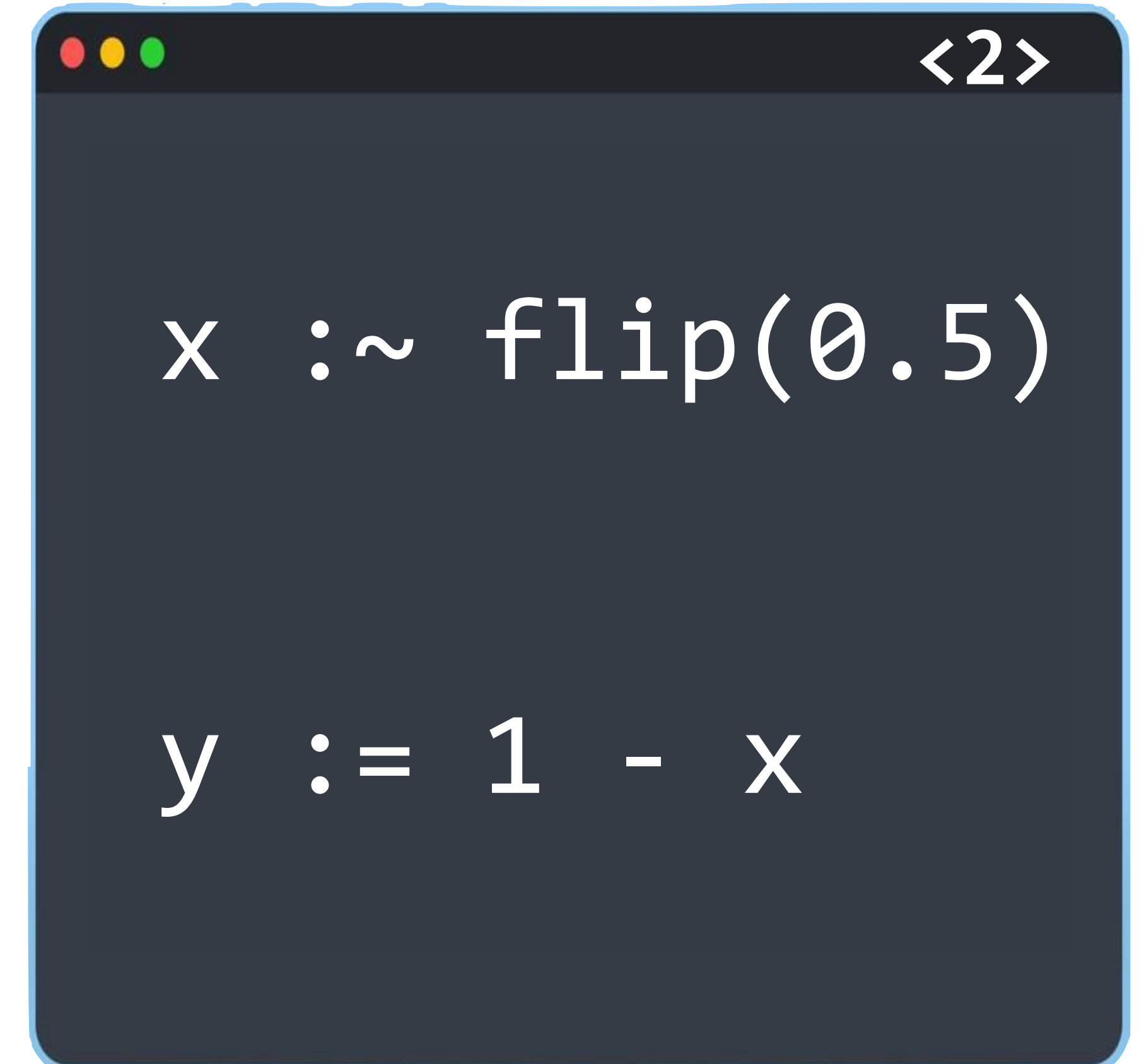
```
<2>  
  
y :~ flip(0.7)  
  
x :~ flip(0.5)
```

$\nVdash [x\langle 1 \rangle = x\langle 2 \rangle \wedge y\langle 1 \rangle = y\langle 2 \rangle]$

Counter-example



```
<1>  
  
x := flip(0.5)  
  
y := x
```



```
<2>  
  
x := flip(0.5)  
  
y := 1 - x
```

$$\llbracket x\langle 1 \rangle = x\langle 2 \rangle \rrbracket \wedge \llbracket y\langle 1 \rangle = y\langle 2 \rangle \rrbracket \not\models \llbracket x\langle 1 \rangle = x\langle 2 \rangle \wedge y\langle 1 \rangle = y\langle 2 \rangle \rrbracket$$

Independence Allows us to do so:

RL-MERGE

$$\llbracket R_1 \rrbracket * \llbracket R_2 \rrbracket \vdash \llbracket R_1 \wedge R_2 \rrbracket$$

Unary Reasoning

Relational Reasoning

Bluebell



Unary v.s.
Relational

Why mixing?

How?

Joint Conditioning — Unary Case

$$x \sim \text{flip}(0.5) \quad [y = 1 - x]$$

Another atomic proposition:

$y = 1 - x$ with probability 1.

(Measure-theoretic definitions are used so that the resource needed to assert such ‘certain’ proposition is independent from other parts)

Joint Conditioning — Unary Case

$$x \sim \text{flip}(0.5) * [y = 1 - x]$$

with probability 0.5

$$[x = 0] * [y = 1 - x]$$

with probability 0.5

$$[x = 1] * [y = 1 - x]$$

$$C_{\text{flip}(0.5)}^v \cdot [x = v] * [y = 1 - x]$$

Joint Conditioning — Relational Case

$$x\langle 1 \rangle \sim \text{flip}(0.5) * [y\langle 1 \rangle = 1 - x\langle 1 \rangle]$$

$$x\langle 2 \rangle \sim \text{flip}(0.5) * [y\langle 2 \rangle = x\langle 2 \rangle]$$

with probability 0.5

$$[x\langle 1 \rangle = 0] * [y\langle 1 \rangle = 1 - x\langle 1 \rangle]$$

$$[x\langle 2 \rangle = 0] * [y\langle 2 \rangle = x\langle 2 \rangle]$$

with probability 0.5

$$[x\langle 1 \rangle = 1] * [y\langle 1 \rangle = 1 - x\langle 1 \rangle]$$

$$[x\langle 2 \rangle = 1] * [y\langle 2 \rangle = x\langle 2 \rangle]$$

$$C_{\text{flip}(0.5)}^v \cdot [x\langle 1 \rangle = v] * [x\langle 2 \rangle = v] * [y\langle 1 \rangle = 1 - x\langle 1 \rangle] * [y\langle 2 \rangle = x\langle 2 \rangle]$$

Joint Conditioning — Relational Case

$$x\langle 1 \rangle \sim \text{flip}(0.5) * [y\langle 1 \rangle = 1 - x\langle 1 \rangle]$$

$$x\langle 2 \rangle \sim \text{flip}(0.5) * [y\langle 2 \rangle = x\langle 2 \rangle]$$

with probability 0.5

$$[x\langle 1 \rangle = 0] * [y\langle 1 \rangle = 1 - x\langle 1 \rangle]$$

$$x\langle 2 \rangle \sim \text{flip}(0.5) * [y\langle 2 \rangle = x\langle 2 \rangle]$$

with probability 0.5

$$[x\langle 1 \rangle = 1] * [y\langle 1 \rangle = 1 - x\langle 1 \rangle]$$

$$x\langle 2 \rangle \sim \text{flip}(0.5) * [y\langle 2 \rangle = x\langle 2 \rangle]$$

$$C_{\text{flip}(0.5)}^v \cdot [x\langle 1 \rangle = v] * [y\langle 1 \rangle = 1 - x\langle 1 \rangle] * x\langle 2 \rangle \sim \text{flip}(0.5) * [y\langle 2 \rangle = x\langle 2 \rangle]$$

Joint Conditioning Modality

$f_1 \models C_\mu v . P(v)$ if there exist kernels

κ_1 such that $f_1 = (\mu \gg \kappa_1)$

and for all value v in the support of μ ,

$\kappa_1(v) \models P(v).$

Encoding Conditional Independence

Example.

$$\exists \mu . C_{\mu} v . [x = v] * y \sim \text{flip}(v) * z \sim \text{flip}(v)$$

$$\exists \mu . C_{\mu} v . [x = v] * y \text{ measurable} * z \text{ measurable}$$

**y and z are conditional
independent given x**

Encoding Relational Lifting

Example. $[y\langle 1 \rangle = y\langle 2 \rangle]$ can be encoded as

$$\exists \mu . \mu(a = b) = 1 \wedge C_{\mu}(a, b) . [y\langle 1 \rangle = a] * [y\langle 2 \rangle = b]$$

Joint conditioning satisfies a lot of useful,
expressive rules.

Rules for Joint Conditioning Modality

It commutes with many connectives in the logic.

C-FOR-ALL

$$C_{\mu} v. \forall x : X. Q(v, x) \vdash \forall x : X. C_{\mu} v. Q(v, x)$$

Rules for Joint Conditioning Modality

It inherits rules from monadic laws.

C-UNIT-L

$$C_{\delta_{v_0}} v. K(v) \dashv\vdash K(v_0)$$

C-UNIT-R

$$E\langle i \rangle \sim \mu \dashv\vdash C_\mu v. \lceil E\langle i \rangle = v \rceil$$

C-ASSOC

$$\frac{\mu_0 = \text{bind}(\mu, \lambda v. (\text{bind}(\kappa(v), \lambda w. \text{return}(v, w))))}{C_\mu v. C_{\kappa(v)} w. K(v, w) \vdash C_{\mu_0}(v, w). K(v, w)}$$

From rules for the joint conditioning modality, we can derive

- All rules in Probabilistic Relational Hoare Logic
- New principles for derive relational liftings

RL-MERGE

$$\llbracket R_1 \rrbracket * \llbracket R_2 \rrbracket \vdash \llbracket R_1 \wedge R_2 \rrbracket$$

RL-CONVEX

$$C_\mu _ . \llbracket R \rrbracket \vdash \llbracket R \rrbracket$$

Unary reasoning also
becomes more
powerful!

Example. One-Time Pad

$\{ T \}$

$m : \sim \text{flip}(0.7);$

$\{ m \sim \text{flip}(0.7) \}$

$\{ m \sim \text{flip}(0.7) * T \}$

$k : \sim \text{flip}(0.5);$

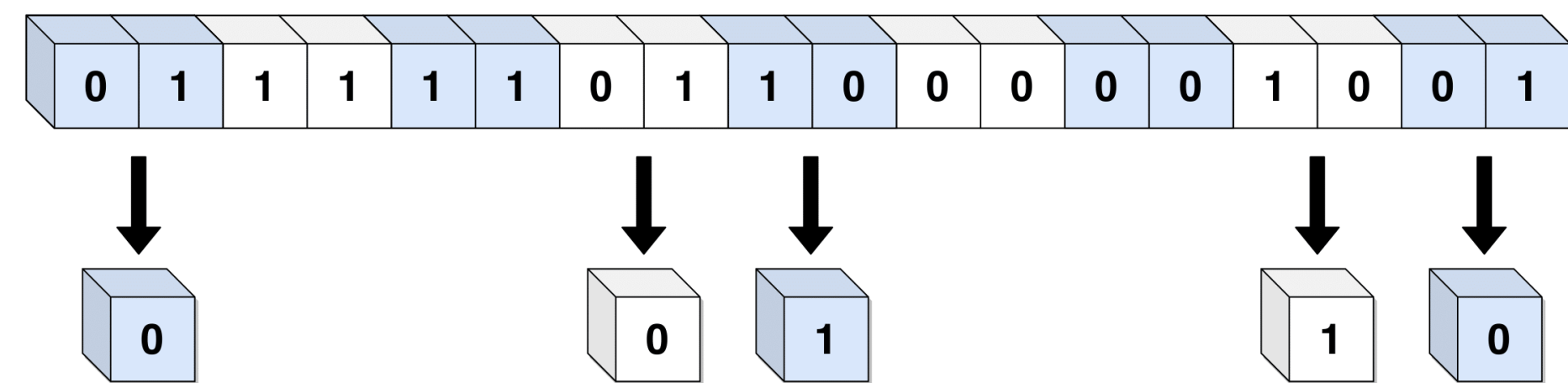
$\{ m \sim \text{flip}(0.7) * k \sim \text{flip}(0.5) \}$

$c : \sim m \text{ xor } k$

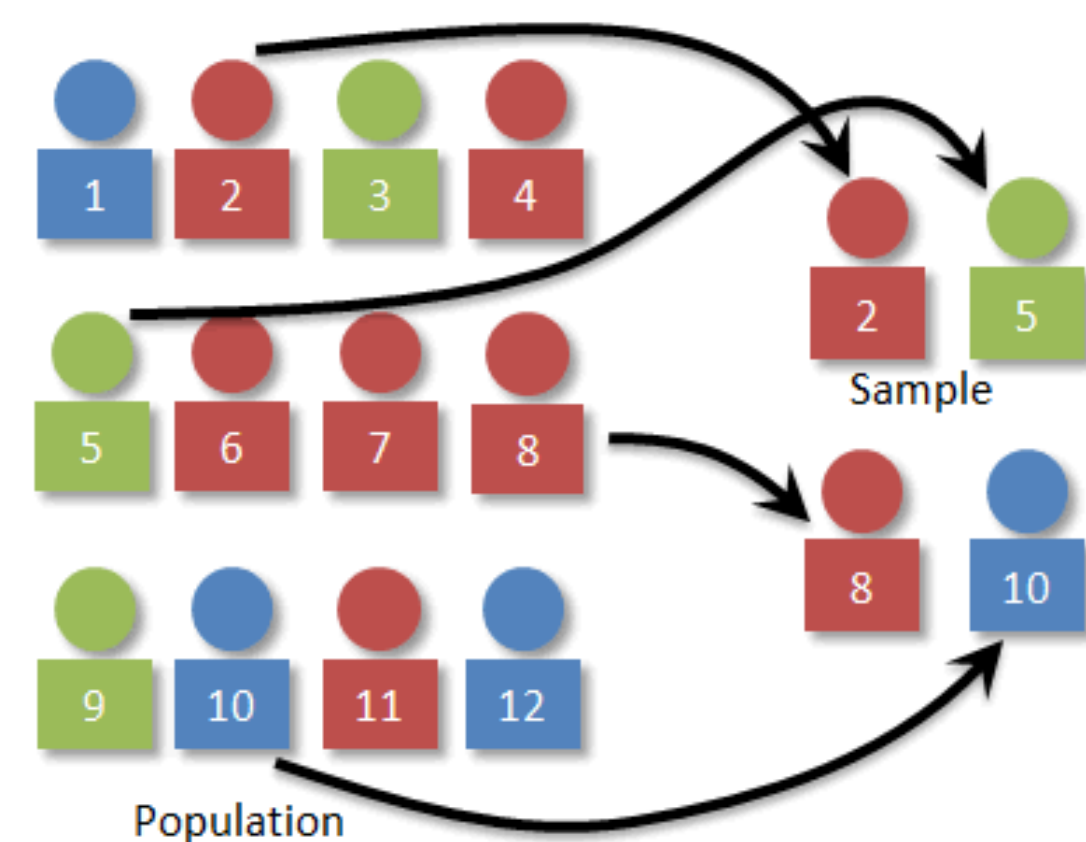
With rules for C_μ , no need for custom axioms here

$\{ m \sim \text{flip}(0.7) * c \sim \text{flip}(0.5) \}$

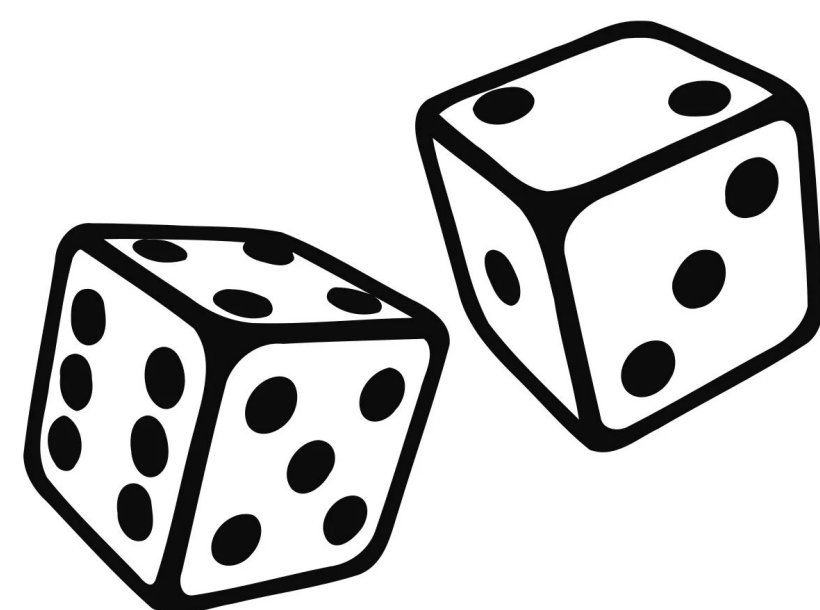
More Examples



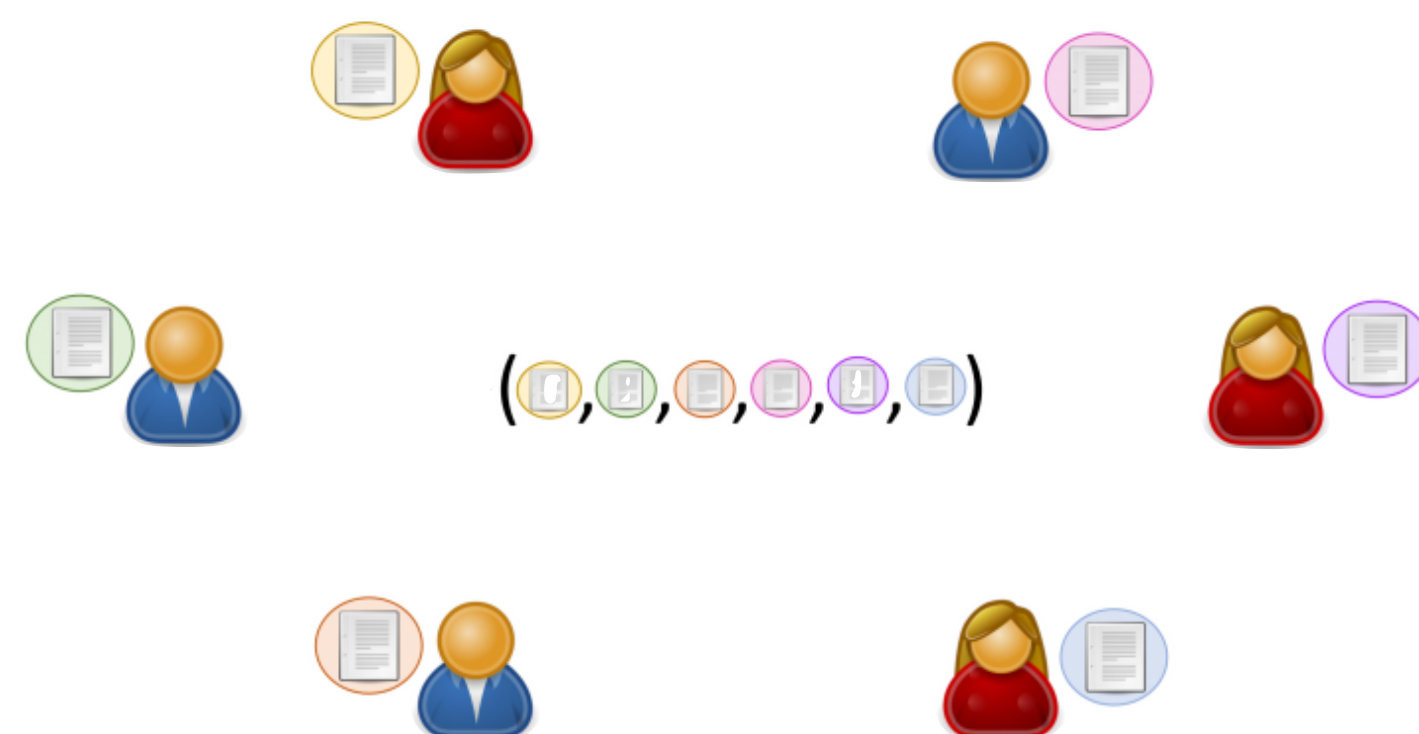
Von Neumann Extractor



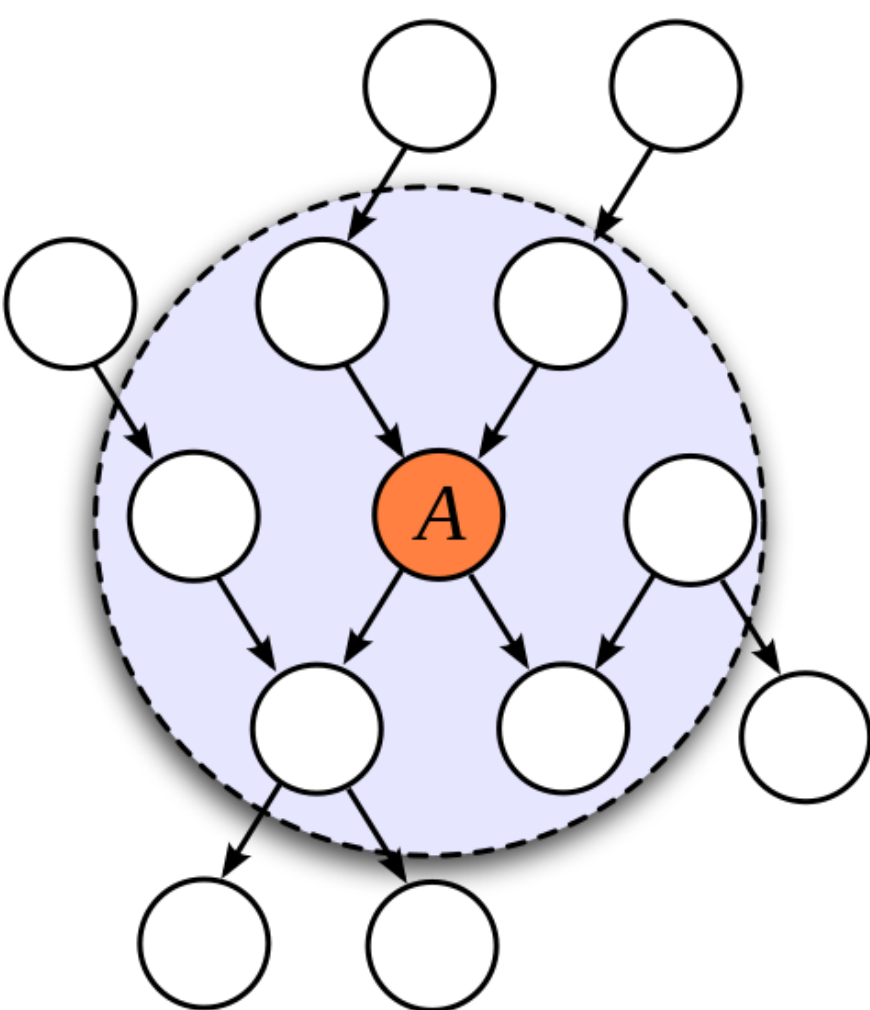
Reservoir Sampling



Monte Carlo Estimation



Multi-party Computation



Markov Blanket

Probabilistic Relational Logic

- **Formal certification of code-based cryptographic proofs.** Barthe, Gregoire and Zanella. POPL 2009

The Coupling Method in Program Logic

- **Relational reasoning via probabilistic coupling.** Barthe, Espitau, Grégoire, Hsu, Stefanescu, and Strub. LPAR 2015
- **Asynchronous Probabilistic Couplings in Higher-Order Separation Logic.** Gregersen, Aguirre, Haselwarter, Tassarotti, Birkedal. POPL 2024
- ...

Probabilistic Independence

- **A Probabilistic Separation Logic.** Barthe, Hsu and Liao. POPL 2020

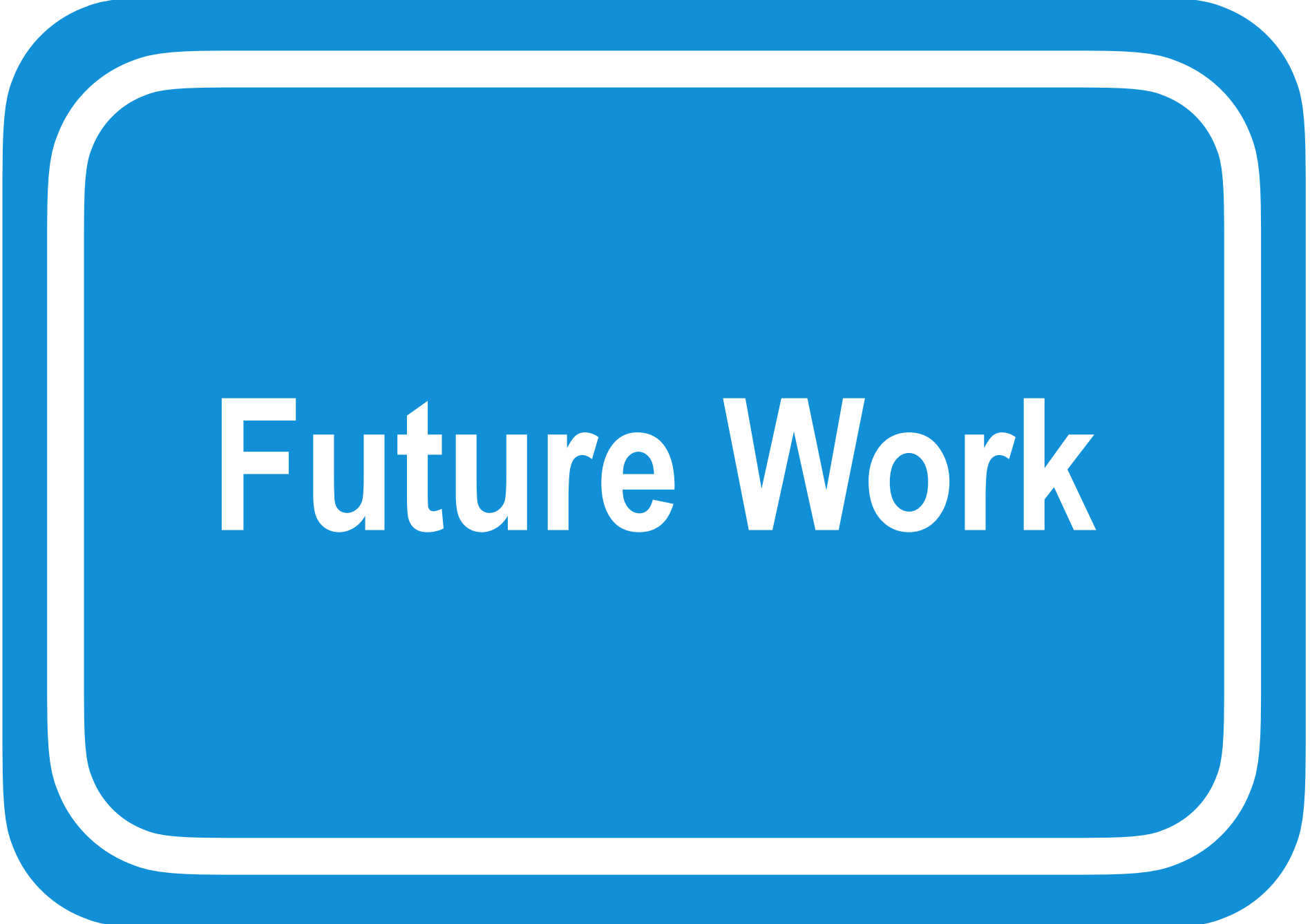
Independence as Separation

- **Lilac: A Modal Separation Logic for Conditional Probability.** Li, Amhed, Holtzen. PLDI 2023

Unary Conditioning Modality

- **A Bunched Logic for Conditional Independence.** Bao, Docherty, Hsu, and Silva. LICS 2021

Conditional Independence



Future Work

Future Work

- A loop rule for unbounded loops
- Automation of the logic
- Supports for continuous probability
- ...



Unary Reasoning
Independence
Conditioning Modality

*

Relational Reasoning:
Relational Lifting

[R]

Bluebell



Joint Conditioning Modality

$C_\mu v . P(v)$

New Proof Patterns

RL-CONVEX

$C_\mu _ . [R] \vdash [R]$

RL-MERGE

$[R_1] * [R_2] \vdash [R_1 \wedge R_2]$